

QF-SPAD V2 Test report

1 Experimental setup:

In the experimental setup, an AeroDIODE CW laser operating at -35 dBm was used, and the laser power was attenuated by 60 dB before being sent to the SPAD. Three RF signals were generated by the ZCU-102 FPGA with an FMC card: a START signal, a GATE signal applied to the QF-SPAD, and a 10 MHz clock signal provided to the Time-Tagger. The SPAD output was connected to Stop 1 of the Time-Tagger. The setup and signal are shown in the Figure 1

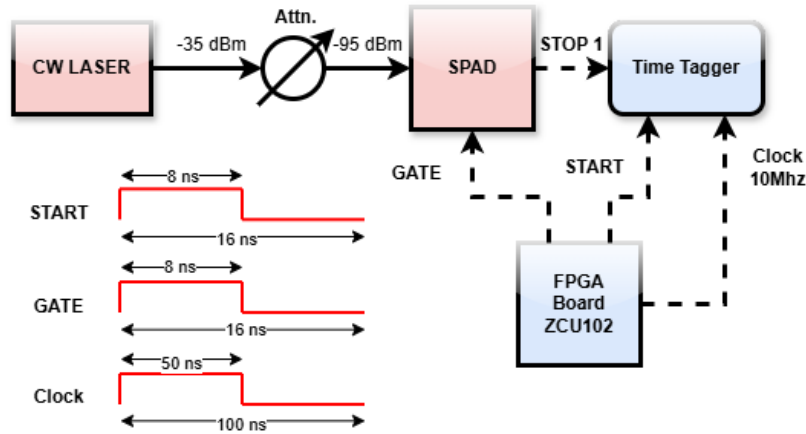


Figure 1: Experimental setup for QF-SPAD-V2

2 Test procedure and Commands Used to run the SPAD - V2

Download the Python code attached to Redmine issue #3997 (this procedure applies only to the QF-SPAD V2). Open PowerShell, navigate to the directory containing the Python script, and run the program. Follow the ON and OFF instructions provided below.

2.1 Turn On commands

- Power up the SPAD using a 5 V, 2 A USB adapter.
- Connect the USB-C cable to the system.
- Run the Python program using PowerShell.
- Select the appropriate COM port.
- SPAD_TEMP=1 - Check the SPAD temperature (by default, it should be between -19 °C and -23 °C).
- TEC_INIT=1 - Initialize the TEC; the temperature will decrease and be set to -39 °C.
- TEMP_INTR=1 - Lock the temperature at -39 °C.
- SPAD_TEMP_READ=1 - Read the current SPAD temperature.

- SPAD_BIAS=160 - Increase the bias voltage (range: 0 to 63.3 V).
- SPAD_BIAS_READ - Monitor the SPAD bias voltage.
- OUTPUT_WIDTH= 4 - Set the output pulse width (range: 0–4, corresponding to 10 ns to 50 ns).
- GATE_SRC=1 - Select the internal gate trigger (use GATE_SRC=0. for an external gate trigger).
- GATE_FREQ=10 - Set the gate frequency to 10 MHz (not required when using an external gate).
- GATE_WIDTH=511 - Set the gate width (range: 127–511, corresponding to approximately 5.1 ns).
- GATE_OUT_LEVEL=214 - Control the level shifter amplitude; a value of 214 corresponds to 6 V and is considered a safe operating level.
- DEAD_TIMR=2 - Set the dead time (range: 0–2, corresponding to 5 μ s to 15 μ s).
- Finally, use SPAD_TEMP_READ=1 and SPAD_BIAS_READ to continuously monitor the SPAD temperature and bias voltage.

2.2 Turn Off commands and procedure

- Remove the external gate RF connector.
- SPAD_BIAS=0 - Set the bias voltage to 0 and wait until it decreases from 160 to 0.
- SPAD_BIAS_READ - Verify that the bias voltage is close to zero (e.g., 0.052 V).
- Unplug the USB-C connector from the PC.
- Power off the device.

3 Test table

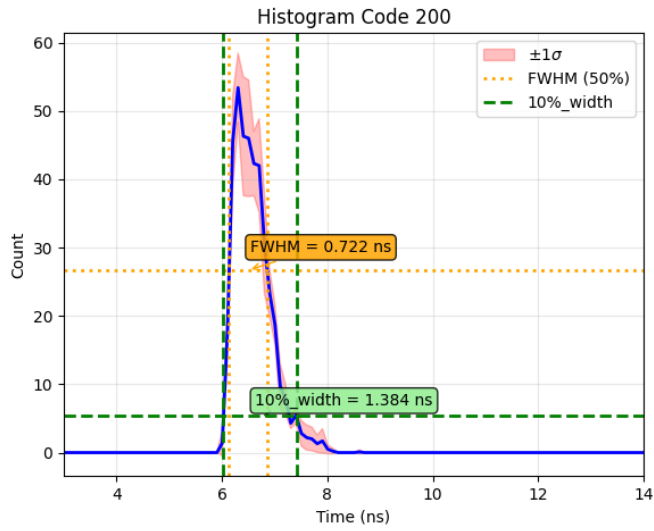
Please click the [Link](#) to see the all test results with the different bias code and it's corresponding voltage. All Bias code are conformed with the Qf- team.

Gate width code	FWHM bias code 196	10% width bias code 196	FWHM bias code 184	10% width bias code 184	Ash-Results FWHM bias code 184	Ash-Results 10% width bias code 184
200	0.722	1.384	0.486	1.214	0.738	1.324
250	1.242	1.859	0.835	1.616	0.556	1.314
300	1.765	2.563	1.334	2.085	0.974	1.751
350	2.464	3.098	1.97	2.769	1.606	2.409
400	3.096	3.72	2.689	3.418	2.241	3.012
450	3.613	4.281	3.163	3.954	2.695	3.483
500	4.111	4.803	3.715	4.506	3.196	3.981
511	4.231	4.842	3.797	4.515	3.294	4.077

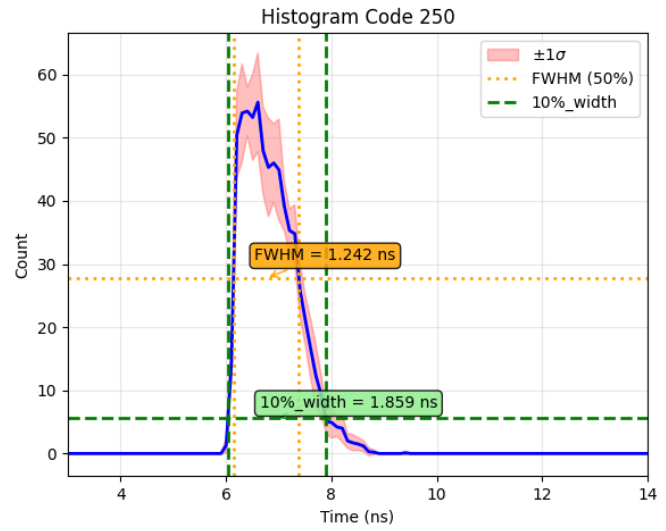
Table 1: Test results.

4 Test results:

The test results of the Bias code of 196 the bias voltage is 68 V. Conformed with the Qf-team to work with this voltage.

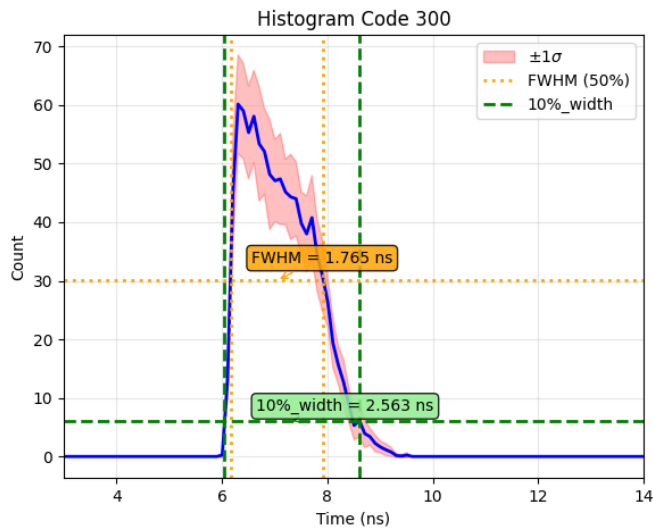


a. Gate width 200

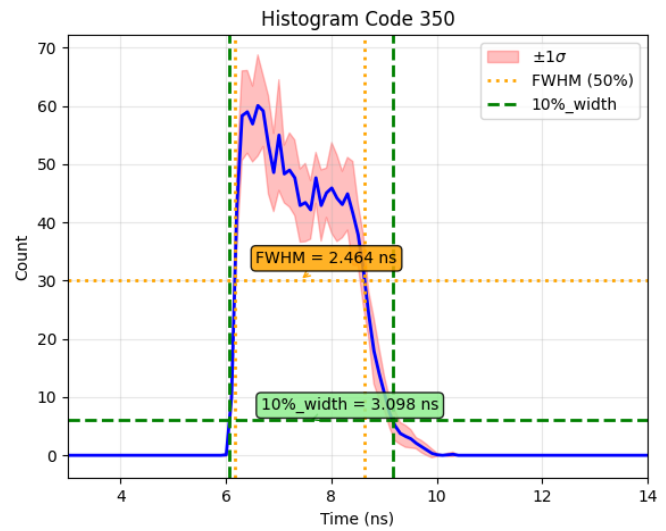


b. Gate width 250

Figure 2

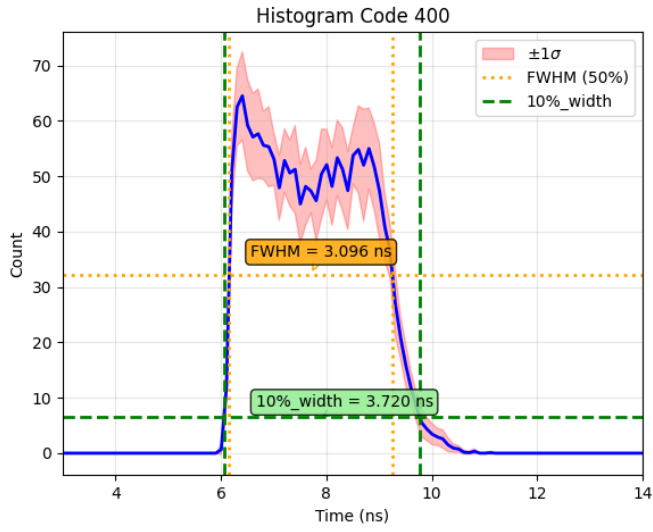


a. Gate width 300

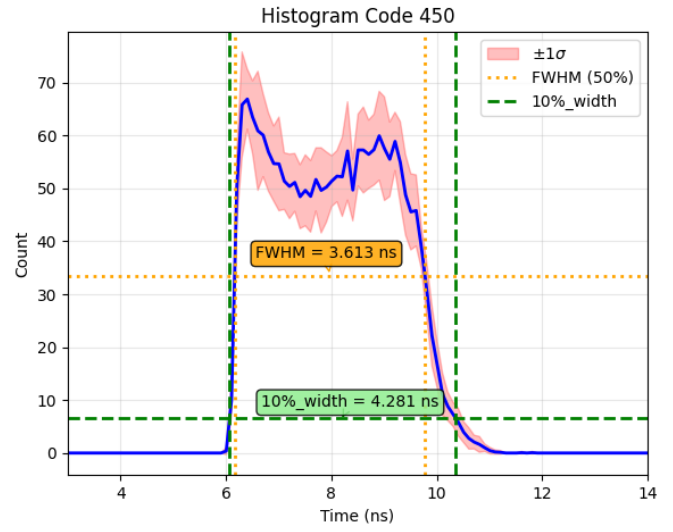


b. Gate width 350

Figure 3

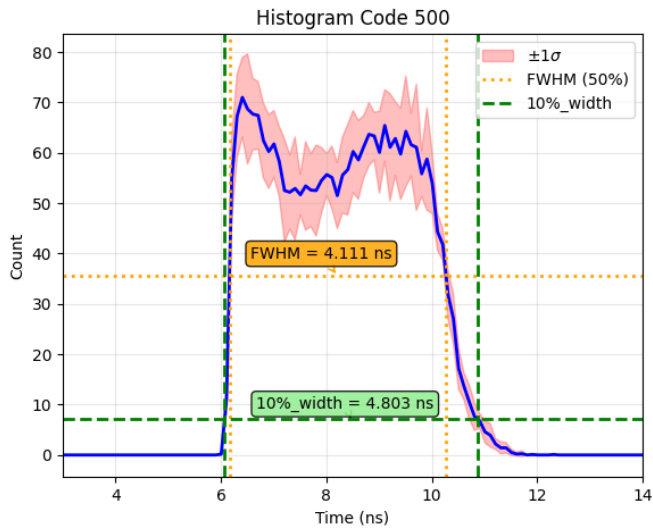


a. Gate width 400

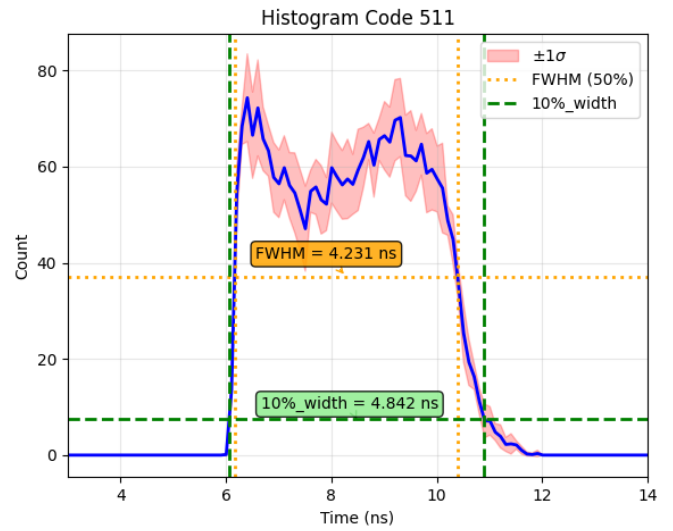


b. Gate width 450

Figure 4



a. Gate width 500



b. Gate width 511

Figure 5